

Case Study: Windows 11 Laptop Performance Optimization

Overview

A customer reported that their HP laptop was experiencing severe performance issues, including slow application launches, delayed system responsiveness, and poor overall usability during normal daily tasks.

The objective was to diagnose the root cause of the performance degradation, perform corrective actions where possible, and provide recommendations for long-term improvement.

Device Information

Manufacturer: HP

Model: HP Laptop 14-dq3xxx

Operating System: Windows 11 Home 25H2

Processor: Intel Celeron N4500 @ 1.10 GHz

Installed Memory: 4 GB DDR4-3200 (Samsung)

Storage: 64 GB SSD/eMMC

Initial Findings

System Performance Metrics

Initial Task Manager analysis revealed:

- Memory utilization consistently between 88% and 92%
- CPU utilization remained relatively low during normal operation
- Disk utilization remained low, indicating storage was not the primary bottleneck
- Only 7.84 GB of free storage remained on the system drive

Startup Applications Review

Several non-essential startup applications were identified and reviewed, including:

- HP System Tray

- HP Support components
- Microsoft Edge Startup Processes
- Mobile Device integrations

Unnecessary startup applications were disabled to reduce background resource consumption.

Installed Applications Review

Installed software was evaluated to identify potential performance impacts.

Findings included:

- McAfee Personal Security
- HP Support and Analytics applications
- Microsoft Teams components
- Dropbox Lite
- Roblox Studio and Roblox Player

The software inventory review was used to identify applications contributing to background resource consumption.

Diagnostic Process

Memory Analysis

Task Manager Performance monitoring revealed:

- Installed RAM: 4 GB
- Available Memory: Less than 400 MB
- Memory utilization consistently above 90%

This indicated that the operating system was exhausting available memory resources during normal operation.

Storage Analysis

Storage utilization showed:

- Total Capacity: 57.3 GB
- Free Space Before Cleanup: 7.84 GB

Low available storage was identified as a secondary performance concern.

Hardware Verification

PowerShell diagnostics confirmed:

- HP Laptop 14-dq3xxx
- Single Samsung DDR4-3200 memory module installed
- 4 GB total system memory

Hardware analysis indicated a likely upgrade path to 8 GB RAM.

Windows Update Verification

Windows Update status was reviewed.

Results:

- System fully updated
 - No critical updates pending
 - Optional preview update available but not installed
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Corrective Actions Performed

Startup Optimization

- Reviewed startup applications
- Disabled unnecessary startup items
- Reduced background resource consumption

Storage Cleanup

Windows Storage cleanup tools were used to remove:

- Temporary files
- Cached data
- Unnecessary system-generated files

Results:

- Free Storage Before Cleanup: 7.84 GB
- Free Storage After Cleanup: 13.0 GB

Total Storage Recovered:

- Approximately 5.2 GB

Software Assessment

Reviewed installed applications for:

- Security software conflicts
- Unnecessary background services
- Vendor utilities consuming resources

System Health Verification

Performed system diagnostics including:

- Windows Update verification
 - System file integrity review
 - Resource utilization analysis
 - Hardware inventory verification
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Root Cause Analysis

The primary cause of the performance issue was identified as insufficient system memory.

Evidence:

- Memory utilization consistently exceeded 90%
- No abnormal CPU utilization patterns observed
- Storage device functioning normally
- No evidence of storage bottlenecks
- Windows 11 operating system consuming most available memory during standard operation

Secondary contributing factors:

- Low available storage space
 - Background security and vendor utility software
 - Limited hardware resources associated with an entry-level processor platform
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Recommendations

Immediate Recommendation

Upgrade system memory from:

- 4 GB RAM → 8 GB RAM minimum

Expected Benefits:

- Improved multitasking performance
- Reduced memory saturation
- Faster application responsiveness
- Improved overall user experience

Additional Recommendations

- Remove unnecessary vendor utilities where appropriate
 - Remove unused applications
 - Maintain at least 15–20% free disk space
 - Perform periodic storage cleanup
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Outcome

Following optimization activities:

- Recovered over 5 GB of storage capacity
- Reduced unnecessary startup activity
- Verified system update compliance
- Identified the primary hardware bottleneck
- Developed a documented upgrade path

The customer was provided with a clear explanation of the root cause and actionable recommendations for improving long-term system performance.

Skills Demonstrated

- Windows 11 Administration
- Endpoint Troubleshooting
- Performance Diagnostics
- Resource Utilization Analysis
- Hardware Assessment
- Startup Optimization
- Storage Management
- Customer Support
- Root Cause Analysis
- Technical Documentation
- PowerShell Diagnostics
- Service Reporting